

Final exam

- Explain clearly. *Do not use the Jordan form unless the question mentions it.* Otherwise you may use any definition/theorem we did in class (clearly specify what result you are using), unless the question asks you to prove something standard. Ask me in case of any doubt.
 - Upload your answers on moodle. *Separate pdf/jpeg file for each problem.*
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1. Consider the given matrix M with entries in $\mathbb{F}_7$. Find a matrix P for which $P^{-1}MP$ is a diagonal matrix. How many such matrices P are there?

2. (a) Fill in the blanks without proofs. There is a bijection between the sets $\{f \mid f : \mathbb{R}^c \rightarrow \mathbb{R}^r \text{ is a linear map}\}$ and $\{M \mid M \text{ is matrix of real numbers of size } \dots\}$, given by the following recipe.

The matrix M associated to a map f is constructed as follows:

The linear map f associated to a matrix M is defined by $f(v) =$

(b) Consider the given matrix A with given RREF R . Find a basis for each of the following three spaces. Justify briefly by quoting relevant standard results. Your answers should not involve unknown constants in A unless required.

i) the column space of A ii) the row space of A iii) the null space of A .

(c) If possible, find a matrix A with RREF R and RCEF (reduced column echelon form) C . Decide if the following is true or false in general. *"Suppose two matrices X and Y of same size are given. Then a matrix M that is row equivalent to X and column equivalent to Y exists if and only if X and Y have the same rank."*

3. (a) Consider linear operators on $\mathbb{R}^2$ with trace b and determinant 1. For such an operator $T \neq cI$, let $m(T)$ = minimum possible number of nonzero entries in a matrix of T with respect to some basis of $\mathbb{R}^2$. *Give an algorithm to determine $m(T)$ based on the given information about T .*

Carefully formulate your decision procedure and prove its validity, including in particular why a value of $m(T)$ smaller than what you claim is not possible. Clearly organize and specify any cases you make. If at any stage the given information about T is not enough to make a decision, state precisely the minimum further information you need and assume that it is available.

(b) Suppose in part (a) the field $\mathbb{R}$ is replaced by an arbitrary field. Does your answer change? If yes, indicate only the changes you would need to make.

4. Throughout this problem A and B are Hermitian matrices, i.e., $A^* = A$ and $B^* = B$.

(a) Show that (no eigenvalue of A is positive) $\Leftrightarrow (v^*Av \leq 0$ for every column vector v). We say in this case that A is negative semidefinite.

(b) Show that if A is negative semidefinite, then BAB is Hermitian and negative semidefinite.

(c) Show that if A is negative semidefinite, then AB has real eigenvalues. Note: This is true even though AB need not be Hermitian.

5. Let T be an orthogonal operator on a Euclidean space V .

(a) If W is a T -invariant subspace of V , show that so is $W^\perp$.

(b) Show that there is a basis of V with respect to which the matrix of T is block diagonal with each block of size 1×1 or 2×2 .

Possible hint: Fix an orthonormal basis of V so it can be regarded as $\mathbb{R}^n$ with an associated matrix M for T . It may be useful to consider the operator on $\mathbb{C}^n$ associated to M and express vectors in $\mathbb{C}^n$ in terms the real and imaginary parts of their coordinates.

(c) Repeat part (b) if T is assumed to be only normal. State only the changes that are needed. You may appeal without proof to a result on normal operators that you were asked to prove earlier.